

Glossary

▼ Relevant source files

Backend/STUDER/Dockerfile

Backend/STUDER/pom.xml

Backend/STUDER/src/main/java/facu/studer/DTOs/messages/ChatListPageResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/messages/DirectMessagePageResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/messages/DirectMessageResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/notification/NotificationResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/notification/UserNotificationResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/user/FriendResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/user/FriendStatusResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/user/FriendsListResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/user/UserPublicResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/DTOs/user/UserSearchPageResponseDTO.java

Backend/STUDER/src/main/java/facu/studer/StuderApplication.java

Backend/STUDER/src/main/java/facu/studer/config/ApplicationStart.java

Backend/STUDER/src/main/java/facu/studer/config/I18nConfig.java

Backend/STUDER/src/main/java/facu/studer/controllers/DirectMessageController.java

Backend/STUDER/src/main/java/facu/studer/controllers/UserController.java

Backend/STUDER/src/main/java/facu/studer/entities/Friend.java

Backend/STUDER/src/main/java/facu/studer/exceptions/GlobalExceptionHandler.java

Backend/STUDER/src/main/java/facu/studer/mappers/UserMapper.java

Backend/STUDER/src/main/java/facu/studer/security/JwtAuthenticationFilter.java

Backend/STUDER/src/main/java/facu/studer/security/JwtUtil.java

Backend/STUDER/src/main/java/facu/studer/security/SecurityConfig.java

Docker/docker-compose.yml

Frontend/STUDER/public/favicon.ico

Frontend/STUDER/src/app/app.component.html

Frontend/STUDER/src/app/app.component.ts

Frontend/STUDER/src/app/app.config.ts

Frontend/STUDER/src/app/app.routes.ts

Frontend/STUDER/src/app/features/home/home.component.ts

Frontend/STUDER/src/app/shared/components/rich-text/entity-popover/entity-popover.component.ts

Frontend/STUDER/src/app/shared/services/rich-text-parser.service.ts

Frontend/STUDER/src/assets/i18n/en.json

Frontend/STUDER/src/assets/i18n/es.json

Frontend/STUDER/src/styles.css

This glossary defines the technical terms, architectural patterns, and domain-specific concepts used throughout the STUDER codebase. It serves as a reference for onboarding engineers to understand the mapping between high-level educational concepts and their implementation in the code.

Domain Concepts

Term	Definition	Implementation / Code Pointer
Rich Text Entity	A tokenized string starting with a specific prefix (<code>@</code> , <code>#</code> , <code>&</code> , <code>\$</code> , <code>%</code>) that triggers special rendering and popover behavior.	RichTextParserService Frontend/STUDER/src/app/shared/services/rich-text-parser.service.ts 4-7
Friendship Model	A bidirectional or unidirectional social link between users, tracking both following status and mutual friendship.	Friend Backend/STUDER/src/main/java/facu/studer/entities/Friend.java

Term	Definition	Implementation / Code Pointer
Notification LinkedType	Enum defining the target entity of a notification (USER, SYSTEM, MESSAGE, EVENT, DISCUSSION).	NotificationResponseDTO Backend/STUDER/src/main/java/facu/studer/DTOs/notification/NotificationResponseDTO.java
Soft Delete	A pattern where records are marked as inactive rather than removed from the database, often used for users and discussions.	UserController Backend/STUDER/src/main/java/facu/studer/controllers/UserController.java

Sources: Frontend/STUDER/src/app/shared/services/rich-text-parser.service.ts 4-7

Backend/STUDER/src/main/java/facu/studer/controllers/UserController.java

Technical Abbreviations & Patterns

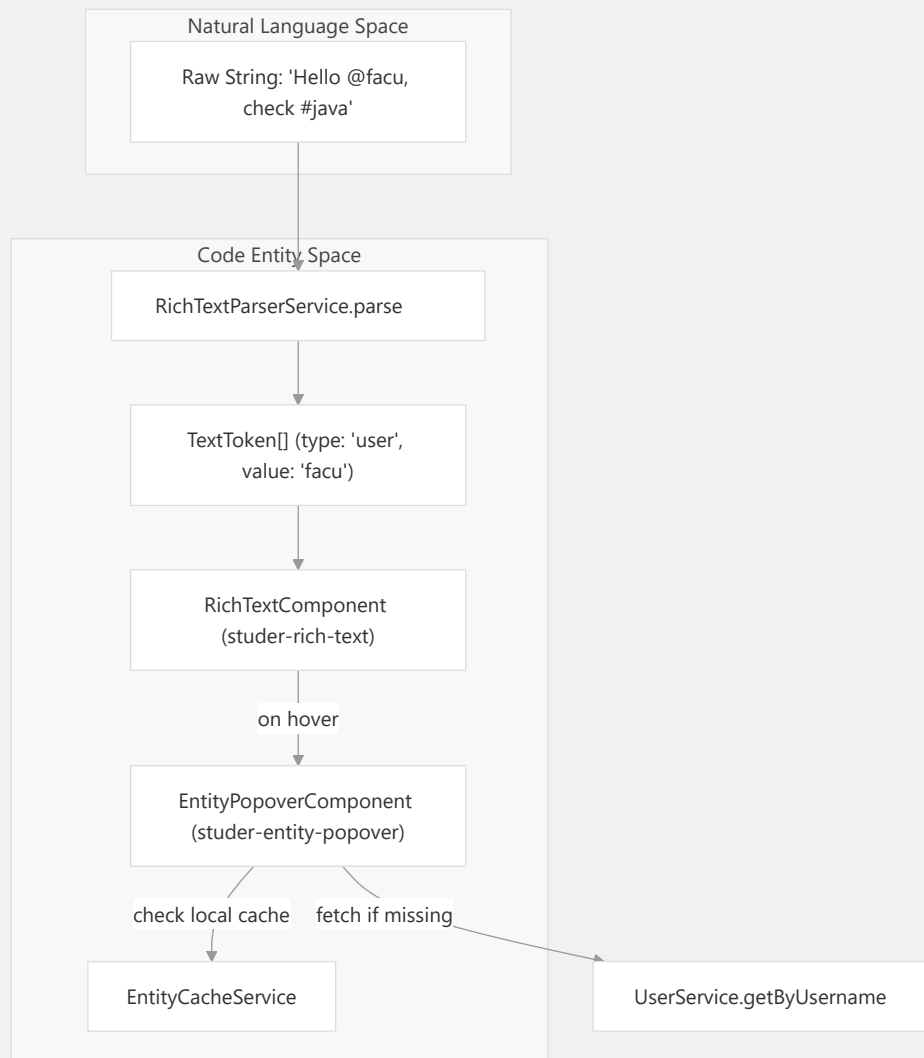
1. Rich Text Tokenization

STUDER uses a custom regex-based tokenizer to identify entities within user-generated content. These entities allow users to link to other profiles, tags, or courses dynamically.

Prefix Mapping:

- **@: User** - Links to a user profile.
- **#: Tag** - Categorizes content.
- **&: Course** - Links to educational modules.
- **\$: Contest** - Links to programming competitions.
- **?: Block** - Links to specific learning units.

Data Flow: From String to Interactive UI The following diagram illustrates how a raw string is processed into interactive components.



Sources: Frontend/STUDER/src/app/shared/services/rich-text-parser.service.ts 30-36

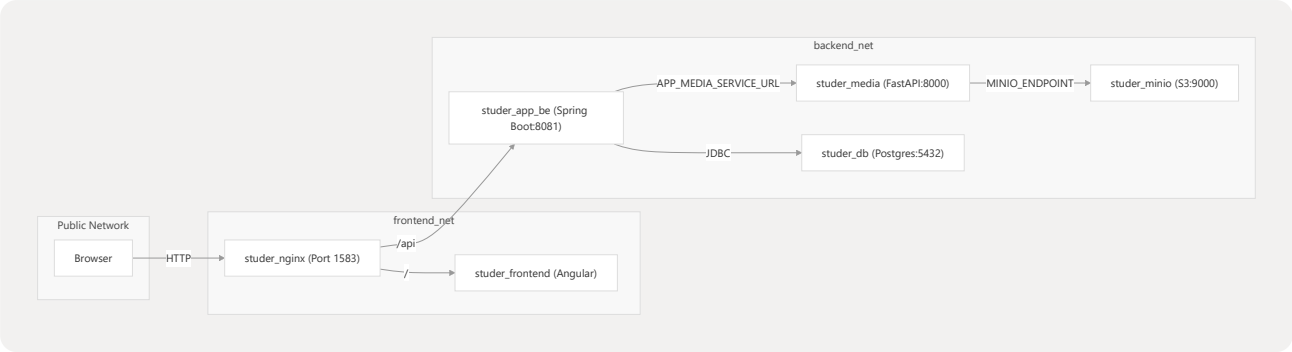
Frontend/STUDER/src/app/shared/components/rich-text/entity-popover/entity-popover.component.ts 60-118

2. Infrastructure & Networking

The system utilizes a multi-container Docker architecture with segmented networks to isolate frontend and backend traffic.

- **frontend_net** : Connects the **nginx** reverse proxy to the **frontend** Angular container.
Docker/docker-compose.yml 91-92
- **backend_net** : Connects **app-be** (Spring Boot), **studer-media** (FastAPI), **db** (PostgreSQL), and **minio** (S3 storage). Docker/docker-compose.yml 93-94

System Component Interaction This diagram bridges the deployment names defined in `docker-compose.yml` with the internal service communication logic.



Sources: `Docker/docker-compose.yml` 2-86 `Backend/STUDER/Dockerfile` 1-14

3. Internationalization (i18n)

STUDER implements i18n on both the frontend and backend to support multi-language environments (English and Spanish).

- **Backend Strategy:** Uses `AcceptHeaderLocaleResolver` to determine the locale based on the `Accept-Language` header sent by the client.

`Backend/STUDER/src/main/java/facu/studer/config/I18nConfig.java` 13-18

- **Frontend Strategy:** Uses `ngx-translate` with JSON assets. Themes and languages are managed via reactive services. `Frontend/STUDER/src/assets/i18n/en.json` 1-10

`Frontend/STUDER/src/assets/i18n/es.json` 1-10

Sources: `Backend/STUDER/src/main/java/facu/studer/config/I18nConfig.java` 13-18

`Frontend/STUDER/src/assets/i18n/en.json` 1-174

Theming System

The application uses CSS variables defined in `:root` to handle runtime theme switching between `theme-light` and `theme-dark`.

Variable	Light Mode Value	Dark Mode Value
<code>--color-primary</code>	<code>#4285f4</code>	<code>#4285f4</code>
<code>--color-bg</code>	<code>#fff</code>	<code>#181818</code>

Variable	Light Mode Value	Dark Mode Value
--color-text-prim	#171717	#f5f5f5
--sidebar-bg	#f4f4f5	#1e1e22

Sources: Frontend/STUDER/src/styles.css 36-143

Project Metadata

- **Main Application Class:** facu.studer.StuderApplication
Backend/STUDER/src/main/java/facu/studer/StuderApplication.java 7
- **Build System:** Maven (Backend), Angular CLI (Frontend).
- **Java Version:** 17 Backend/STUDER/pom.xml 30
- **Spring Boot Version:** 4.0.6 (Parent POM) Backend/STUDER/pom.xml 8

Sources: Backend/STUDER/src/main/java/facu/studer/StuderApplication.java 7-15

Backend/STUDER/pom.xml 5-31